



April 22, 2025

Connie Ellis
Compliance and Reporting Section
Kansas Department of Health & Environment
1000 SW Jackson, Suite 310
Topeka, KS 66612

**RE: Evergy, Inc. – Lawrence Energy Center (Source ID No. 0450014)
MATS Compliance Reports**

Evergy, Inc. is providing the compliance reports for our Lawrence Energy Center MATs affected units – Units 4 and 5.

If there are any questions about this information, please contact me at (785) 418-5542 or craig.adkinson@evergy.com

Sincerely,

A handwritten signature in black ink that reads 'Craig T. Adkinson'.

Craig Adkinson
Evergy, Inc.
Environmental Consultant

Compliance Certification

National Emission Standards for Hazardous Air Pollutants:
Coal and Oil Fired Electric Utility Steam Generating Units
40 CFR Part 63 Subpart UUUUU

Reporting Period: January 1, 2025 – March 31, 2025

Company Name: Evergy, Inc.

Mailing Address: 818 South Kansas, PO Box 889

City: Topeka State: Kansas Zip: 66601

Facility Name: Lawrence Energy Center Source ID: 0450014

Address: 1250 North 1800 Rd

City: Lawrence State: Kansas Zip: 66044

Hazardous Air Pollutants monitored: Mercury

LEC unit 5 – 4,318 MMBtu/hr coal-fired EGU				
Pollutant	Emission Limit	Monitoring equipment manufacturer	Monitoring equipment serial #	Date of latest CMS certification or audit
PM	0.03 lb/MMBtu	N/A	N/A	
SO ₂	0.2 lb/MMBtu	Teledyne	M100 E	8/1/24
Hg	1.2 lb/TBtu	N/A	N/A	1/05/24 - 2/10/24

LEC unit 4 North – 1,409 MMBtu/hr coal-fired EGU				
Pollutant	Emission Limit	Monitoring equipment manufacturer	Monitoring equipment serial #	Date of latest CMS certification or audit
PM	0.03 lb/MMBtu	N/A	N/A	
SO ₂	0.2 lb/MMBtu	Teledyne	M100 E	7/30/24
Hg	1.2 lb/TBtu	N/A	N/A	7/5/- 8/16/2024

LEC unit 4 South – 1,409 MMBtu/hr coal-fired EGU				
Pollutant	Emission Limit	Monitoring equipment manufacturer	Monitoring equipment serial #	Date of latest CMS certification or audit
PM	0.03 lb/MMBtu	N/A	N/A	
SO ₂	0.2 lb/MMBtu	Teledyne	M100 E	7/30/24
Hg	1.2 lb/TBtu	N/A	N/A	7/5/- 8/16/24

Report the total fuel use by each affected source subject to an emission limit, for each calendar month within the semiannual reporting period, including, but not limited to, a description of the fuel, whether the fuel has received a non-waste determination by EPA or your basis for concluding that the fuel is not a waste, has any fuel received a non-waste determination by EPA if not, include the basis for including fuel is not a waste:

The only fuels burned are coal and natural gas. No waste fuel is used.

Did you burn new fuel types during the reporting period?

No.

Include the date of the most recent tune-up for each EGU. The date of the tune-up is the date the tune-up provisions specified in §63.10021(e)(6) and (7) were completed.

Unit	Date of most recent tune-up
LEC Unit 5	02/24/22
LEC Unit 4	05/09/23

*the tune-ups were conducted as specified in 63.10021(e)(6) and (7)

Report the which definition of startup the units rely on.

All units rely on the “startup” definition provided in paragraph (1) of the definition.

Emergency bypass information shall be reported annually (LEE units only).

No units have emergency bypass capabilities.

Summary of performance test results, include documentation of any operating limits that were reestablished during the tests. (include tables for performance test info for each unit including each emission limits).

There were no operational changes since the last stack test that could increase emissions on either of the three units.

Performance test results are provided in the Attachment.

Total operating time

Please see Attachment.

Emission data summary

Please see Attachment.

CMS performance summary

Please see Attachment.

Description of any changes in CMS, processes or controls.

There were no changes to the CMS, processes or controls.

Were there any deviations from any emission limit, work practice standard, or operating limit during the reporting period? If you have a deviation from any emission limit, work practice standard, or operating limit, you must also submit a brief description of the deviation, the duration of the deviation, emissions point identification, and the cause of the deviation. For each excess emissions occurring at an affected source where you are using a CMS to comply with that emission limit or operating limit, you must include the information required in §63.10(e)(3)(v) in the compliance report specified in section (c).

There were no deviations or excess emissions from the emission limit during the reporting period. There were deviations for monitoring downtime according to 40 CFR 63.10031(c)(9). See attached.

I certify under the penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on information and belief formed after reasonable inquiry, including the person or persons who manage the system, or those persons directly responsible for gathering the information, the stated information in this document is true, accurate, and complete.

Name of Responsible Official: Daniel R. Wilkus

Title: Director – Environmental Services

Signature: 

Digitally signed by Daniel R
Wilkus
Date: 2025.04.25 09:09:13
-05'00'

Date: 04/25/2025

Average Data

Plant: Lawrence Energy Center

Interval: 1 Month

Type: Block

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

Source	STACK4N		STACK4S	
	CHEATIN (MMBTU/HR)	GHEATIN (MMBTU/HR)	CHEATIN (MMBTU/HR)	GHEATIN (MMBTU/HR)
Parameter				
Unit				
01/25	296,689.9	8,221.9	222,923.1	4,854.4
02/25	198,133.6	10,768.3	119,392.7	5,361.9
03/25	182,662.2	17,241.3	135,224.3	10,278.2
Average	225,828.6	12,077.2	159,180.0	6,831.5
Minimum	182,662.2	8,221.9	119,392.7	4,854.4
Maximum	296,689.9	17,241.3	222,923.1	10,278.2
Summation	677,485.7	36,231.5	477,540.1	20,494.5
Geometric Mean	220,615.8	11,514.1	153,248.4	6,443.5
Included Data Points	3	3	3	3
Total number of Data Points	3	3	3	3

F = Unit Offline

E = Exceedance

C = Calibration

S = Substituted

I = Invalid

M = Maintenance

T = Out Of Control

*** = Suspect**

U = Startup

D = Shutdown

Average Data

Plant: Lawrence Energy Center

Interval: 1 Month

Type: Block

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

Source	BOILERS	
Parameter	CHEATIN (MMBTU/HR)	GHEATIN (MMBTU/HR)
Unit		

01/25	0.0I	0.0I
02/25	0.0I	0.0I
03/25	0.0I	0.0I

Average	0.0	0.0
Minimum	0.0	0.0
Maximum	0.0	0.0
Summation	0.0	0.0
Geometric Mean		
Included Data Points	0	0
Total number of Data Points	3	3

F = Unit Offline

E = Exceedance

C = Calibration

S = Substituted

I = Invalid

M = Maintenance

T = Out Of Control

*** = Suspect**

U = Startup

D = Shutdown

Operating Time Duration

Plant: Lawrence Energy Center
Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59
Time Online Criteria: 1 minute(s)

Source: STACK4N (STACK 4 NORTH)

Parameter: MATSON (QA MATS Time Online)

Interval: 001H

Incident ID	Start Date/Time	End Date/Time	Duration (hours)
1 *	01/01/2025 00:00	01/26/2025 23:59	624.00
2	01/28/2025 03:00	01/30/2025 13:59	59.00
3	02/03/2025 23:00	02/05/2025 23:59	49.00
4	02/08/2025 22:00	02/22/2025 18:59	333.00
5	02/24/2025 07:00	02/26/2025 03:59	45.00
6	02/27/2025 07:00	03/01/2025 18:59	60.00
7	03/05/2025 03:00	03/06/2025 16:59	38.00
8	03/08/2025 03:00	03/13/2025 12:59	130.00
9	03/15/2025 16:00	03/17/2025 05:59	38.00
10	03/20/2025 03:00	03/22/2025 09:59	55.00
11	03/23/2025 13:00	03/26/2025 17:59	77.00
12	03/30/2025 06:00	03/31/2025 20:59	39.00

Number of Events: 12
Total Duration: 1,547.00 hours

* Indicates duration incident could have additional data prior to the start date or following the end date.

Operating Time Duration

Plant: Lawrence Energy Center
Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59
Time Online Criteria: 1 minute(s)

Source: STACK4S (STACK 4 SOUTH)

Parameter: MATSON (QA MATS Time Online)

Interval: 001H

Incident ID	Start Date/Time	End Date/Time	Duration (hours)
1 *	01/01/2025 00:00	01/26/2025 23:59	624.00
2	01/28/2025 03:00	01/30/2025 13:59	59.00
3	02/03/2025 23:00	02/05/2025 23:59	49.00
4	02/08/2025 22:00	02/22/2025 18:59	333.00
5	02/24/2025 07:00	02/26/2025 03:59	45.00
6	02/27/2025 07:00	03/01/2025 18:59	60.00
7	03/05/2025 03:00	03/06/2025 16:59	38.00
8	03/08/2025 03:00	03/13/2025 12:59	130.00
9	03/15/2025 16:00	03/17/2025 05:59	38.00
10	03/20/2025 03:00	03/22/2025 09:59	55.00
11	03/23/2025 13:00	03/26/2025 17:59	77.00
12	03/30/2025 06:00	03/31/2025 20:59	39.00

Number of Events: 12
Total Duration: 1,547.00 hours

* Indicates duration incident could have additional data prior to the start date or following the end date.

Operating Time Duration

Plant: Lawrence Energy Center

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

No Operating Time found during the report period for:

Source(s): BOILER5

Parameter(s): MATSON

Interval: 001H

* Indicates duration incident could have additional data prior to the start date or following the end date.

MATS Performance Summary

Company:	Lawrence Energy Center	Report Period:	01/01/2025 00:00 Through 03/31/2025 23:59
Address:	1250 N. 180 Road Lawrence, KS 66044	Total Source Time in Report Period:	1618.92 Hours
Source:	STACK4S	Manufacturer:	TELEDYNE ML
Pollutant:	SO2#MM30 (30 Day Averages for MATS SO2#MM) LB/MMBTU	Model Number:	M100 E
Interval:	30 Day	Cert. Date:	11/08/12 00:00
Limit:			

Causes of CEMS Excess Emissions	Number of Excess Emissions (Hours)	Percent of Operating Time
Control Equipment Problems	0	0.00
Cleaning / Soot Blowing	0	0.00
Other Known Causes	0	0.00
Unknown Causes	0	0.00
Process Problems	0	0.00
Startup / Shutdown	0	0.00
Total duration of excess emission	0.00	0.00

Causes of CEMS Downtime	Duration of Downtime (Hours)	Percent of Operating Time
Monitor Equipment Malfunction	0.00	0.00
Other Known Causes	0.00	0.00
Unknown Causes	0.00	0.00
QA Calibration	0.00	0.00
Non-Monitor Equipment Malfunction	0.00	0.00
Total duration of downtime	0.00	0.00

CMS Downtime Deviation Events

Plant: Lawrence Energy Center

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

No Downtime found during the report period for:

Source (s): STACK4S

Parameter(s): SO2#MM30

Reason Code(s): 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 66, 67, 68, 69, 70, 71, 72, 73, 74

* Indicates duration incident could have additional data prior to the start date or following the end date.

MATS Performance Summary

Company: Lawrence Energy Center
Address: 1250 N. 180 Road
Lawrence, KS 66044
Source: STACK4N
Pollutant: SO2#MM30 (30 Day Averages for MATS
SO2#MM) LB/MMBTU
Interval: 30 Day
Limit:

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59
Total Source Time in Report Period: 1618.92 Hours
Manufacturer: TELEDYNE ML
Model Number: M100 E
Cert. Date: 11/08/12 00:00

Causes of CEMS Excess Emissions	Number of Excess Emissions (Hours)	Percent of Operating Time
Control Equipment Problems	0	0.00
Cleaning / Soot Blowing	0	0.00
Other Known Causes	0	0.00
Unknown Causes	0	0.00
Process Problems	0	0.00
Startup / Shutdown	0	0.00
Total duration of excess emission	0.00	0.00

Causes of CEMS Downtime	Duration of Downtime (Hours)	Percent of Operating Time
Monitor Equipment Malfunction	6.00	0.37
Other Known Causes	0.00	0.00
Unknown Causes	0.00	0.00
QA Calibration	0.00	0.00
Non-Monitor Equipment Malfunction	0.00	0.00
Total duration of downtime	6.00	0.37

CMS Downtime Deviation Events

Plant: Lawrence Energy Center

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

Source: STACK4N

Parameter: SO2#MM30

Interval: 001H

Incident ID	Start Date/Time	End Date/Time	Duration (hours)	Reason Code - Description Action Code - Description
1	01/28/2025 03:00	01/28/2025 08:59	6.00	10 - Monitor Equipment Malfunction 11 - MATS Deviation

Number of Events: 1

Total Duration: 6.00 hours

* Indicates duration incident could have additional data prior to the start date or following the end date.

MATS Performance Summary

Company: Lawrence Energy Center
Address: 1250 N. 180 Road
Lawrence, KS 66044
Source: BOILER5
Pollutant: SO2#MM30 (30 Day Averages for MATS
SO2#MM) LB/MMBTU
Interval: 30 Day
Limit:

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59
Total Source Time in Report Period: 0.00 Hours
Manufacturer: TELEDYNE ML
Model Number: M100 E
Cert. Date: 06/22/18 11:00

Causes of CEMS Excess Emissions	Number of Excess Emissions (Hours)	Percent of Operating Time
Control Equipment Problems	0	0.00
Cleaning / Soot Blowing	0	0.00
Other Known Causes	0	0.00
Unknown Causes	0	0.00
Process Problems	0	0.00
Startup / Shutdown	0	0.00
Total duration of excess emission	0.00	0.00

Causes of CEMS Downtime	Duration of Downtime (Hours)	Percent of Operating Time
Monitor Equipment Malfunction	0.00	0.00
Other Known Causes	0.00	0.00
Unknown Causes	0.00	0.00
QA Calibration	0.00	0.00
Non-Monitor Equipment Malfunction	0.00	0.00
Total duration of downtime	0.00	0.00

CMS Downtime Deviation Events

Plant: Lawrence Energy Center

Report Period: 01/01/2025 00:00 Through 03/31/2025 23:59

Time Online Criteria: 1 minute(s)

No Downtime found during the report period for:

Source (s): BOILERS

Parameter(s): SO2#MM30

Reason Code(s): 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 66, 67, 68, 69, 70, 71, 72, 73, 74

* Indicates duration incident could have additional data prior to the start date or following the end date.



EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

UNIT 4N

Field Data

	Run #1	Run #2	Run #3	Averages
Date:	5/3/2022	5/3/2022	5/3/2022	
Start time:	9:55	12:20	14:35	
Stop time:	12:04	14:34	16:44	

Filterable PM (lb/MMBtu):	0.0063	0.0035	0.0015	0.00379
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Permit Limit
0.030

PASS

MATS LEE Status	
0.015	YES

UNIT 4S

Field Data

	Run #1	Run #2	Run #3	Averages
Date:	5/4/2022	5/4/2022	5/4/2022	
Start time:	9:10	11:40	14:20	
Stop time:	11:17	13:53	16:39	

Filterable PM (lb/MMBtu):	0.0050	0.0028	0.0046	0.00414
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Permit Limit
0.030

PASS

MATS LEE Status	
0.015	YES



EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data								
	Run #1	Run #2	Run #3	Averages	Permit Limit		MATS LEE Status	
Date:	6/2/2022	6/2/2022	6/2/2022		0.030	PASS	0.015	YES
Start time:	8:50	11:35	14:25					
Stop time:	11:05	13:50	16:40					
Filterable PM (lb/MMBtu):	0.0010	0.0011	0.0011	0.00107				

Plant and Unit	Lawrence Energy Center Unit 4N
City & State	Lawrence, KS
Date	7/17 - 8/16/2024
Operator	BJ, SB, AI



Summary - RM Data

	Run #	1	2	3	4	5
	Start Date	07/17/24	07/23/24	07/29/24	08/04/24	08/10/24
	End Date	07/23/24	07/29/24	08/04/24	08/10/24	08/16/24
	Start Time	11:02	12:12	10:11	8:05	7:55
	End Time	11:39	7:50	7:41	7:36	11:05
Average						
Moisture (%)	12.99	12.24	13.13	13.51	12.94	13.10
Sample Volume A (dsl)	1705.48	1735.500	1675.695	1628.326	1721.974	1765.900
Hg mass A (ng)	379.00	188.6	225.8	375.9	688.9	415.8
Breakthrough A (%)	1%	0.1%	4.6%	0.3%	0.4%	0.4%
Hg Concentration A (µg/wscm)	0.19	0.10	0.12	0.20	0.35	0.20
Sample Volume B (dsl)	1705.48	1735.500	1675.694	1628.323	1721.973	1765.900
Hg mass B (ng)	400.42	204.2	244.2	408.0	702.5	443.2
Breakthrough B (%)	1%	1.6%	2.5%	1.7%	0.4%	0.2%
Hg Concentration B (µg/wscm)	0.21	0.10	0.13	0.22	0.36	0.22
Relative Deviation (%)	3%	4.0%	3.9%	4.1%	1.0%	3.2%
Average Hg (µg/wscm)	0.198	0.10	0.12	0.21	0.35	0.21
Average Hg (µg/dscm)	0.228	0.11	0.14	0.24	0.40	0.24
Average lb/Tbtu	0.244	0.12	0.15	0.26	0.43	0.26
Average lb/yr (PTE)	9.229	4.54	5.67	9.83	16.27	9.83

Summary - CEMS Data

	Run #	1	2	3	4	5
Average						
Load (GMW)	73.5	79	76	77	67	69
CO ₂ (%vw)	9.5	9.60	9.70	9.60	9.50	9.30
CO ₂ (%vd)	11.0	10.94	11.17	11.10	10.91	10.70
Days Per Run	6.00	6.00	6.00	6.00	6.00	6.00

Plant and Unit	Lawrence Energy Center Unit 4S
City & State	Lawrence, KS
Date	7/5 - 8/4/2024
Operator	BJ, SB, AI



Summary - RM Data						
	Run #	1	2	3	4	5
	Start Date	07/05/24	07/11/24	07/17/24	07/23/24	07/29/24
	End Date	07/11/24	07/17/24	07/23/24	07/29/24	08/04/24
	Start Time	9:51	8:11	11:12	12:17	10:16
	End Time	7:46	10:07	11:17	7:55	7:37
Average						
Moisture (%)	12.88	12.92	12.65	12.61	13.14	13.09
Sample Volume A (dsl)	1709.418	1703.180	1751.300	1728.960	1675.664	1687.987
Hg mass A (ng)	396.1	667.7	315.4	286.2	287.3	423.8
Breakthrough A (%)	0.4%	0.4%	0.3%	0.0%	1.4%	0.0%
Hg Concentration A (µg/wscm)	0.20	0.34	0.16	0.14	0.15	0.22
Sample Volume B (dsl)	1709.415	1703.173	1751.300	1728.970	1675.657	1687.974
Hg mass B (ng)	396.1	674.7	293.6	278.4	293.6	440.4
Breakthrough B (%)	0.3%	0.4%	0.1%	0.1%	0.8%	0.1%
Hg Concentration B (µg/wscm)	0.20	0.34	0.15	0.14	0.15	0.23
Relative Deviation (%)	1.7%	0.5%	3.6%	1.4%	1.1%	1.9%
Average Hg (µg/wscm)	0.200	0.34	0.15	0.14	0.15	0.22
Average Hg (µg/dscm)	0.232	0.39	0.17	0.16	0.17	0.26
Average lb/Tbtu	0.242	0.41	0.18	0.17	0.18	0.27
Average lb/yr (PTE)	9.154	15.51	6.81	6.43	6.81	10.21
Summary - CEMS Data						
	Run #	1	2	3	4	5
Average						
Load (GMW)	75.2	69	75	79	76	77
CO ₂ (%vw)	9.7	9.60	9.80	9.70	9.80	9.60
CO ₂ (%vd)	11.1	11.02	11.22	11.10	11.28	11.05
Days Per Run	6.00	6.00	6.00	6.00	6.00	6.00

Plant and Unit	Lawrence Energy Center Unit 5
City & State	Lawrence, KS
Date	1/5-2/10/24
Operator	AE/ SB/ AI



Summary - RM Data							
Run #	1	2*	3	4	5	6	Average
Start Date	01/05/24	01/11/24	01/17/24	01/23/24	01/29/24	02/04/24	
End Date	01/11/24	01/17/24	01/23/24	01/29/24	02/04/24	02/10/24	
Start Time	11:10	12:20	14:50	11:21	10:38	13:42	
End Time	12:00	12:52	11:03	10:04	13:09	10:19	
Moisture (%)	14.08	13.60	14.26	14.13	14.07	14.45	14.20
Sample Volume A (dsI)	1737.993	585.069	1682.690	1704.027	1758.302	1687.389	1714.080
Hg mass A (ng)	678.3	343.0	1994.9	933.1	925.4	542.0	1014.7
Breakthrough A (%)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Hg Concentration A (µg/wscm)	0.34	0.51	1.02	0.47	0.45	0.27	0.51
Sample Volume B (dsI)	1737.984	439.511	1682.679	1704.001	1758.289	1687.377	1714.066
Hg mass B (ng)	679.1	403.6	1976.5	936.2	907.9	539.4	1007.8
Breakthrough B (%)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Hg Concentration B (µg/wscm)	0.34	0.79	1.01	0.47	0.44	0.27	0.51
Relative Deviation (%)	0.1%	22.1%	0.5%	0.2%	1.0%	0.2%	0.4%
Average Hg (µg/wscm)	0.34	0.65	1.01	0.47	0.45	0.27	0.508
Average Hg (µg/dscm)	0.39	0.75	1.18	0.55	0.52	0.32	0.592
Average lb/Tbtu	0.36	0.67	1.09	0.51	0.63	0.57	0.632
Average lb/yr (PTE)	13.62	25.34	41.23	19.29	23.83	21.56	23.906

Summary - CEMS Data							
Run #	1	2	3	4	5	6	Average
Load (GMW)	186	290	277	258	153	142	203.1
CO ₂ (% vw)	10.60	11.20	10.70	10.60	8.20	5.50	9.1
CO ₂ (% vd)	12.34	12.96	12.48	12.34	9.54	6.43	10.6
Days Per Run	6.00	6.00	6.00	6.00	6.00	6.00	6.00

* In accordance with KDHE's 1/31/24 approval, this run was deemed invalid and omitted due to ice formations in sample line.